

POST-WAR SUBMARINES

Nuclear-powered submarines and their armament of submarine-launched ballistic missiles (SLBMs) are arguably the most potent weapon system ever developed. Hidden beneath the waves, they are able to launch their deadly missiles without warning and are virtually immune from retaliatory action by the enemy. Their high level of concealment also provides them with an invaluable second-strike capability. More conventional attack submarines, powered either by nuclear reactors or diesel-electric motors, operate against surface vessels or hunt down other submarines.

▼ USS MARYLAND (SSBN-738)

Commissioned 1992 **Origin** US

Displacement 16,000 tons
(18,700 tons submerged)

Length 171m (561ft)

Top speed 20 knots (25 knots submerged)

An *Ohio*-class ballistic missile submarine, *Maryland* is at the forefront of America's nuclear deterrent. Its 24 Trident missiles are capable of delivering up to 12 warheads per missile to a range of 11,300km (7,021 miles).



► NOVOSIBIRSK (B-401)

Commissioned 1984

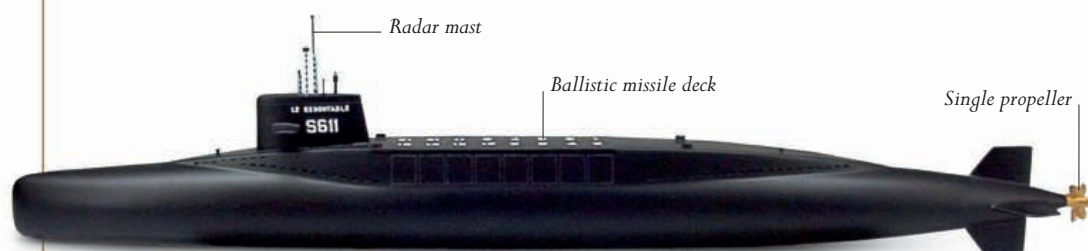
Origin Soviet Union

Displacement 2,325 tons
(3,075 tons submerged)

Length 67m (220ft)

Top speed 15 knots (24 knots submerged)

A diesel-electric attack submarine of the *Kilo* class, *Novosibirsk* was armed with minelaying equipment and conventional torpedoes, as well as anti-ship and anti-aircraft missiles.



◀ LE REDOUTABLE (S611)

Commissioned 1971 **Origin** France

Displacement 8,045 tons
(8,940 tons submerged)

Length 128m (420ft)

Top speed 18 knots (25 knots submerged)

Le Redoutable was the lead vessel in its class and France's first ballistic missile submarine. Its original M1 missiles had a range of 2,000km (1,243 miles) while carrying a 450 kilotonne warhead.



▲ AKULA CLASS

Commissioned 1981

Origin Soviet Union

Displacement 18,500 tons
(25,000 tons submerged)

Length 175m (574ft)

Top speed 22 knots (27 knots submerged)

Codenamed Typhoon by NATO, the Russian *Akula* ("shark") class of submarines were the largest undersea vessels ever built. They were designed to fire SLBMs while hidden under the Arctic ice.

Shrouded propeller
to minimize sound



▲ USS NAUTILUS (SSN-571)

Commissioned 1954 **Origin** US

Displacement 3,535 tons
(4,090 tons submerged)

Length 98.8m (324ft)

Top speed 22 knots (25 knots submerged)

A ground-breaking vessel, the *Nautilus* was the world's first operational nuclear-powered submarine and, in 1958, the first to complete an underwater transit of the North Pole.

Missile hatch



Bow fitted with
six torpedo tubes